

1. Lock-gate equations

Write the Boolean equations for each output of the lock gate you are implementing.

- Identify the inputs and outputs.
- Write each output as a Boolean expression.
- Verify the equations using a truth table.
- Make sure the equations match the required reversible logic gate.

2. Wire order

Define the order of the inputs and outputs in your circuit.

For example, if your gate has three inputs and three outputs:

$$(A, B, C) \rightarrow (P, Q, R)$$



- Input order: A, B, C
- Output order: P, Q, R

Keep the same order in your truth tables, VHDL port declarations, and testbenches.

3. Useful outputs

Identify which outputs perform the intended logical operation and which outputs are additional outputs required to preserve reversibility.

For example, a full-adder circuit may produce:

$$\text{Sum} = A \oplus B \oplus C_{in}$$

$$C_{out} = AB + AC_{in} + BC_{in}$$

The remaining outputs depend on the reversible gate being used.

4. Ancilla and garbage definitions

Ancilla inputs: Additional inputs set to a constant, usually 0, to help implement a desired function.

Garbage outputs: Outputs that are not needed for the primary function but must be retained to preserve the reversible mapping.

Example:

$$(A, B, 0) \rightarrow (P, Q, R)$$



Here, the third input is an ancilla input initialized to zero. Whether an output is garbage depends on the function and the circuit's requirements.

5. Depth rule

Define the logic depth of your circuit.

- Logic depth is the number of sequential gate levels along a path from an input to an output.
- Gates operating in parallel count as one level.
- Lower depth can reduce circuit delay.

For example, if a circuit has three sequential gate levels, its logic depth is 3.

6. Quantum cost (QC) convention

Define how you will calculate the quantum cost of each reversible gate and the complete circuit.

$$QC_{\text{total}} = \sum_{i=1}^n QC_i$$


Use the cost values specified by your instructor or course materials. Do not assume that every reversible gate has the same quantum cost.
